

Tide retrospective:

anharmonic_id_gibbsRegularity skeleton (2/3 fields proven)

Claude (Opus 4.7, 1M context)

2026-05-23

Abstract

A skeleton tide producing the file `Laplace/OneD/AnharmonicGibbsRegularity.lean` that supplies the 2/3 easy fields of the `Threepoint.GibbsRegularity` instance for the 1D anharmonic potential at the linear perturbation $A(x) = x$. `partition_pos` is proven via `integral_pos_iff_support_of_nonneg_ae` + the (newly public) `integrable_exp_neg_t_anharmonic` lemma. `partition_h_zero` closes by `congr`; `funext`; `ring_nf`. `partition_hasDerivAt` is left as a `sorry` with a detailed proof sketch in the docstring; the follow-up analytic-regularity tide picks it up via the dominated-convergence + Gaussian-decay machinery in `Laplace.OneD.IntegralRemainder`.

1 Setting

This tide is the direct handoff from the 2026-05-23 `fdt-identity-1d` tide-experiment¹, which empirically validated the FDT identity $\partial_h \text{Cov}_h(x, x)|_{h=0} = -t \cdot \kappa_3(x, x, x)$ on the 1D anharmonic Gibbs. The general FDT identity is already formalised as `Threepoint.gibbsCov_deriv_eq_neg_t_kappa3` in the `Threepoint` package; the remaining work to apply it to the anharmonic case is the `GibbsRegularity` instance for the anharmonic potential.

The harmonic mirror `Threepoint.harmonic_id_gibbsRegularity` (in `HarmonicGibbsRegularity.lean`, Tide 13/14) routes through the closed-form Gaussian partition function. The anharmonic case has no elementary closed form for $Z(h)$, so the `partition_hasDerivAt` field is genuinely harder and needs dominated convergence with explicit Gaussian-decay bounds. This skeleton tide lands the easy 2/3 fields; the hard third is deferred.

2 What landed

New file: `Laplace/OneD/AnharmonicGibbsRegularity.lean` (155 lines), exporting:

- `integrable_exp_neg_t_anharmonic` — public witness for $\exp(-t \cdot L_{\text{anh}}) \in L^1(\mathbb{R})$ under the discriminant condition. Proven via the $\exp(-t \cdot L_{\text{anh}}) \leq \exp(-t \cdot c \cdot x^2)$ domination (using `anharmonic_coercive`). Mirrors the $n = 0$ case of the private `AnharmonicKappa3Affine.integrable_pow_mul_e` but exposed publicly so the regularity instance can cite it.
- `anharmonic_partition_pos` — strict positivity of the unperturbed anharmonic partition. Routes through `MeasureTheory.integral_pos_iff_support_of_nonneg_ae`: the integrand is positive everywhere (`Real.exp_ne_zero`), so its support is all of $\mathbb{R}$, which has infinite Lebesgue measure.
- `Threepoint.anharmonic_id_gibbsRegularity` — the `GibbsRegularity` instance with:

¹2026-05-23-experiment-fdt-identity-1d.

- `partition_pos := anharmonic_partition_pos hlam hgamma hdisc ht`
- `partition_h_zero := by congr 1; funext x; ring_nf`
- `partition_hasDerivAt := sorry` (with detailed proof sketch in the docstring; 100–200 lines estimated for the follow-up tide).

Build status: clean (`lake build Laplace.OneD.AnharmonicGibbsRegularity` succeeds in ~ 5 s; one `sorry` warning, no errors).

3 Why a skeleton, not the full instance

Per the `laplace CLAUDE.md` proof-estimate rule: probability of direct one-shot proof on `partition_hasDerivAt` estimated at $\sim 30\%$, well below the 50% threshold for “factor into intermediate lemmas first”. The hard field needs:

- A dominator on the perturbed integrand $\exp(-t(L_{\text{anh}}(x) + h \cdot x))$ for $|h| \leq 1$, valid for the h -derivative-under-integral lemma. The dominator is $\exp(-t \cdot L_{\text{anh}}(x)) \cdot \exp(t|x|) \cdot (|tx| + 1)$, Gaussian-decaying after completing the square via `anharmonic_coercive`.
- An invocation of `MeasureTheory.hasDerivAt_integral_of_dominated_loc_of_lip` or its newer Mathlib variant.
- Final pointwise computation of the $h = 0$ derivative inside the integral.

Estimated 100–200 lines. Left as a focused follow-up tide.

4 Why this still matters

With the skeleton in place:

- The instance type-checks and is callable from any downstream module (modulo the `sorry`). This means dependent tides can be drafted against the instance signature *now*, not blocked on the full proof.
- The two easy fields are real proofs (not sorries): they exercise the `integral_pos_iff_support_of_nonneg_ae + coercivity` pattern, which the next tide will need to extend to the h -perturbed case.
- The follow-up tide has a sharp scope: only `partition_hasDerivAt`, with all the imports, namespace setup, and the surrounding lemmas already in place.

5 Pipeline lessons

- **Sanity-check Mathlib API before relying on names.** My first attempt cited `volume_univ_eq_top`, a name that doesn’t exist. The correct Mathlib name is `Real.volume_univ` (in the `Real` namespace, defined in `Mathlib.MeasureTheory.Measure.Lebesgue.Basic`). A `grep -rn` on the actual Mathlib source resolved it in ~ 30 seconds.
- **Build the skeleton before attempting the full proof.** The 2/3 easy fields built in one shot; the hard third was left as a documented `sorry`. This pattern is preferable to trying the full instance and risking thrashing on the hard part.
- **Public-ify private lemmas at the boundary.** The `integrable_pow_mul_exp_neg_anharmonic` lemma in `AnharmonicKappa3Affine.lean` was `private`; the $n = 0$ case is needed by this file, so it was re-derived (essentially copy-paste) as the public `integrable_exp_neg_t_anharmonic`. A future cleanup tide could promote the original to public and remove the redundancy; for now, the two coexist.

6 Follow-ups

- **partition_hasDerivAt follow-up tide.** ~ 100 – 200 lines. Dominated convergence + Gaussian-decay domination via `anharmonic_coercive`. The full proof sketch is in the docstring of the `sorry` site.
- **GibbsObservable instances at $\phi \in \{x, x^2, x^3, x \cdot x, x^2 \cdot x, x \cdot x \cdot x\}$.** Six instances, each ~ 30 – 50 lines after the `partition_hasDerivAt` pattern is in place. These discharge the per-observable hypotheses of `gibbsCov_deriv_eq_neg_t_kappa3`. Total ~ 200 lines.
- **Headline corollary** (once both above land): `cov_h_id_id_deriv_anharmonic_eq_neg_t_kappa3`, ~ 20 – 30 lines, by direct application of the general Threepoint theorem.
- **Closed-form asymptotic corollary:** compose the headline with Tide 9 (`kappa3_anharmonic_id_id_id_asymp`) to get $\lim_{t \rightarrow \infty} t \cdot \partial_h \text{Cov}_h|_{h=0} = \alpha/\lambda^3$. ~ 30 lines.

Acknowledgements

Direct handoff from the same-day `2026-05-23-experiment-fdt-identity-1d` tide-experiment which surfaced the formalisation gap. The original tide proposal in `projects/primer/tide-log/2026-05-23-tide-anhar` mis-identified the gap as “general FDT identity”; the revised proposal (after discovering `Threepoint.gibbsCov_deriv` is already proven) correctly identified the gap as “anharmonic regularity instance”. This tide takes the first ($\sim 30\%$) bite of that revised target.